3.57 Verify Stokes's theorem for the vector field

$$\mathbf{B} = (\hat{\mathbf{r}}\cos\phi + \hat{\boldsymbol{\phi}}\sin\phi)$$

by evaluating:

(a) $\oint_C \mathbf{B}\cdot d\boldsymbol{\ell}$ over the path comprising a quarter section of a circle, as shown in **Fig. P3.57**, and

(b) $\int_S (\nabla\times\mathbf{B})\cdot d\mathbf{s}$ over the surface of the quarter section.

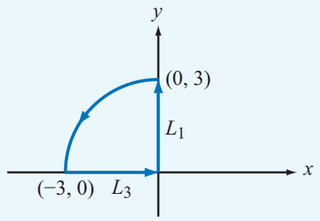


Figure P3.57 Problem 3.57.

3.58 Determine if each of the following vector fields is solenoidal, conservative, or both:

*(a) $\mathbf{A} = \hat{\mathbf{x}}x^2 - \hat{\mathbf{y}}2xy$

(b) $\mathbf{B} = \hat{\mathbf{x}}x^2 - \hat{\mathbf{y}}y^2 + \hat{\mathbf{z}}2z$

(c) $\mathbf{C} = \hat{\mathbf{r}}(\sin\phi)/r^2 + \hat{\boldsymbol{\phi}}(\cos\phi)/r^2$

*(d) $\mathbf{D} = \hat{\mathbf{R}}/R$

(e) $\mathbf{E} = \hat{\mathbf{r}}\left(3 - \frac{r}{1+r}\right) + \hat{\mathbf{z}}z$

(f) $\mathbf{F} = (\hat{\mathbf{x}}y + \hat{\mathbf{y}}x)/(x^2+y^2)$

(g) $\mathbf{G} = \hat{\mathbf{x}}(x^2+z^2) - \hat{\mathbf{y}}(y^2+x^2) - \hat{\mathbf{z}}(y^2+z^2)$

*(h) $\mathbf{H} = \hat{\mathbf{R}}(Re^{-R})$

3.59 Find the Laplacian of the following scalar functions:

(a) $V = 4xy^2z^3$

(b) $V = xy + yz + zx$

*(c) $V = 3/(x^2+y^2)$

(d) $V = 5e^{-r}\cos\phi$

(e) $V = 10e^{-R}\sin\theta$

3.60 Find the Laplacian of the following scalar functions:

(a) $V_1 = 10r^3\sin 2\phi$

(b) $V_2 = (2/R^2)\cos\theta\sin\phi$

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3.57 验证向量场的斯托克斯定理

$$\mathbf{B} = (\hat{\mathbf{r}}\cos\phi + \hat{\boldsymbol{\phi}}\sin\phi)$$

通过评估：

(a) 在包含四分之一圆截面的路径上的 $\oint_C \mathrm{B}\cdot\mathrm{d}\boldsymbol{\ell}$，如图 P3.57 所示，以及 (b) 在四分之一截面的表面上的 $\int_S (\nabla\times\mathrm{B})\cdot\mathrm{ds}$。

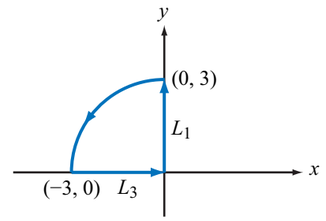


图 P3.57 问题 3.57。

3.58 确定以下每个矢量场是螺线管矢量场、保守矢量场还是两者兼而有之：

*(a) $\mathbf{A} = \hat{\mathbf{x}}x^2 - \hat{\mathbf{y}}2xy$

(b) $\mathbf{B} = \hat{\mathbf{x}}x^2 - \hat{\mathbf{y}}y^2 + \hat{\mathbf{z}}2z$

(c) $\mathbf{C} = \hat{\mathbf{r}}(\sin\phi)/r^2 + \hat{\boldsymbol{\phi}}(\cos\phi)/r^2$

*(d) $\mathbf{D} = \hat{\mathbf{R}}/R$

(e) $\mathbf{E} = \hat{\mathbf{r}}\left(3 - {}_{1+r}^{r}\right) + \hat{\mathbf{z}}z$

(f) $\mathbf{F} = (\hat{\mathbf{x}}y + \hat{\mathbf{y}}x)/(x^2+y^2)$

(g) $\mathbf{G} = \hat{\mathbf{x}}(x^2+z^2) - \hat{\mathbf{y}}(y^2+x^2) - \hat{\mathbf{z}}(y^2+z^2)$

*(h) $\mathbf{H} = \hat{\mathbf{R}}(Re^{-R})$

3.59 求下列标量函数的拉普拉斯算子：

(a) $\mathrm{V} = 4\mathrm{xy}^2\mathrm{z}^3$b) V

=(xy + yz + zx *(c)

$\mathrm{V} = 3/(\mathrm{x}^2+\mathrm{y}^2)$d) V

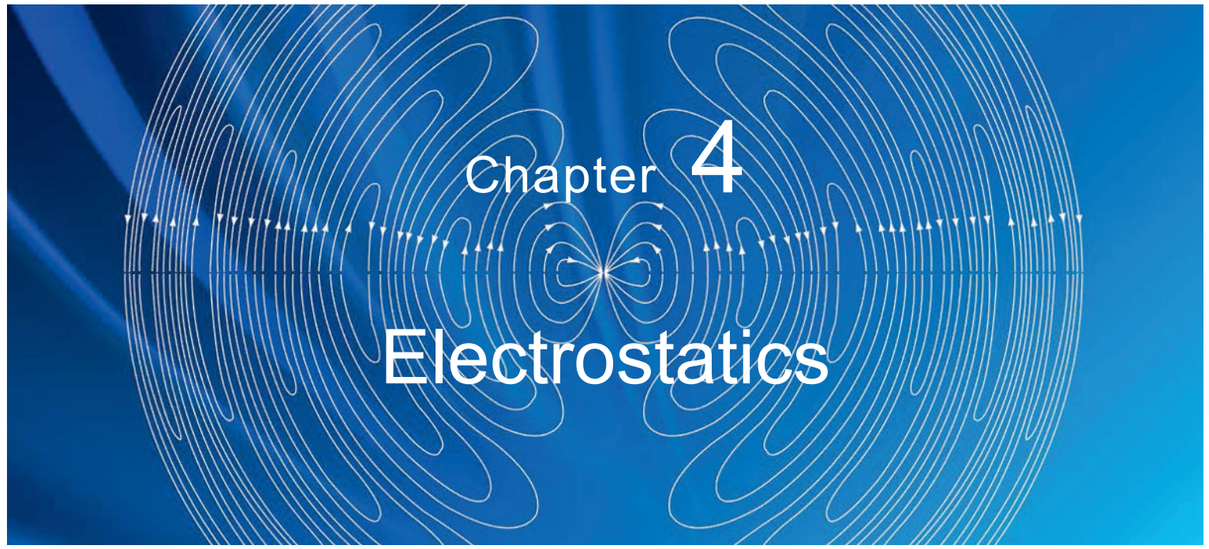
$= 5\mathrm{e}^{-r}\cos\phi$ e) V = 1

$0\mathrm{e}^{-R}$ 罪恶 θ

3.60 求下列标量函数的拉普拉斯算子：

(a) $\mathrm{V}_1 = 10\mathrm{r}^3\sin 2\phi$ b) $\mathrm{V}_2 = (2/\mathrm{R}^2)\cos\theta\sin\phi$

(



Chapter 4 Electrostatics

第4章 静电场

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Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Evaluate the electric field and electric potential due to any distribution of electric charges.
2. Apply Gauss's law.
3. Calculate the resistance R of any shaped object given the electric field at every point in its volume.
4. Describe the operational principles of resistive and capacitive sensors.
5. Calculate the capacitance of two-conductor configurations.

章节内容

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目标

学习本章介绍的材料后，您应该能够：

1. 评估由于电荷分布而产生的电场和电势。 2. 应用高斯定律。 3. 在给定其体积内每个点的电场的情况下，计算任何形状物体的电阻 R。 4. 描述电阻式和电容式传感器的工作原理。 5. 计算双导体配置的电容。

4-1 Maxwell's Equations

The modern theory of electromagnetism is based on a set of four fundamental relations known as ***Maxwell's equations***:

$$\nabla \cdot \mathbf{D} = \rho_v, \tag{4.1a}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \tag{4.1b}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{4.1c}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}. \tag{4.1d}$$

Here **E** and **D** are the ***electric field intensity*** and ***flux density*** interrelated by $\mathbf{D} = \varepsilon\mathbf{E}$ where ε is the electrical permittivity; **H** and **B** are ***magnetic field intensity*** and ***flux density*** interrelated by $\mathbf{B} = \mu\mathbf{H}$ where μ is the magnetic permeability; ρ_v is the electric charge density per unit volume; and **J** is the current density per unit area. The fields and fluxes **E**, **D**, **B**, **H** were introduced in Section 1-3, and ρ_v and **J** will be discussed in Section 4-2. Maxwell's equations hold in any material, including free space (vacuum). In general, all of the above quantities may depend on spatial location and time t. In the interest of readability, we will not, however, explicitly reference these dependencies (as in $\mathbf{E}(x,y,z,t)$) except when the context calls for it. By formulating these equations, published in a classic treatise in 1873, James Clerk Maxwell established the first unified theory of electricity and magnetism. His equations are deduced from experimental observations reported by Coulomb, Gauss, Ampère, Faraday, and others; they not only encapsulate the connection between the electric field and electric charge and between the magnetic field and electric current but also capture the bilateral coupling between electric and magnetic fields and fluxes. Together with some auxiliary relations, Maxwell's equations comprise the fundamental tenets of electromagnetic theory.

► Under ***static*** conditions, none of the quantities appearing in Maxwell's equations are functions of time (i.e., $\partial/\partial t = 0$). *This happens when all charges are permanently fixed in space. If they move, they do so at a steady rate so that ρ_v and* **J** *are constant in time.* ◄

Under these circumstances, the time derivatives of **B** and **D** in Eqs. (4.1b) and (4.1d) vanish, and Maxwell's equations reduce to the following pairs.

Electrostatics

$$\nabla \cdot \mathbf{D} = \rho_v, \tag{4.2a}$$

$$\nabla \times \mathbf{E} = 0. \tag{4.2b}$$

Magnetostatics

$$\nabla \cdot \mathbf{B} = 0, \tag{4.3a}$$

$$\nabla \times \mathbf{H} = \mathbf{J}. \tag{4.3b}$$

Maxwell's four equations separate into two uncoupled pairs with the first pair involving only the electric field and flux **E** and **D** and the second pair involving only the magnetic field and flux **H** and **B**.

► Electric and magnetic fields become ***decoupled*** under static conditions. ◄

This allows us to study electricity and magnetism as two distinct and separate phenomena as long as the spatial distributions of charge and current flow remain constant in time. We refer to the study of electric and magnetic phenomena under static conditions as ***electrostatics*** and ***magnetostatics***, respectively. Electrostatics is the subject of the present chapter, and we learn about magnetostatics in Chapter 5. The experience gained through studying electrostatic and magnetostatic phenomena will prove invaluable in tackling the more involved material in subsequent chapters that deal with time-varying fields, charge densities, and currents.

We study electrostatics not only as a prelude to the study of time-varying fields but also because it is an important field in its own right. Many electronic devices and systems are based on the principles of electrostatics. They include x-ray machines, oscilloscopes, ink-jet electrostatic printers, liquid crystal displays, copy machines, micro-electromechanical switches and accelerometers, and many solid-state–based control devices. Electrostatic principles also guide the design of medical diagnostic sensors, such as the electrocardiogram, which records the heart's pumping pattern, and the electroencephalogram, which records brain activity, as well as the development of numerous industrial applications.

4-2 Charge and Current Distributions

In electromagnetics, we encounter various forms of electric charge distributions. When put in motion, these charge distributions constitute current distributions. Charges and currents may be distributed over a volume of space, across a surface, or along a line.

4-1 麦克斯韦方程组

现代电磁理论基于一组称为麦克斯韦方程组的四个基本关系：

$$\nabla \cdot \mathbf{D} = \rho_v, \tag{4.1a}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \tag{4.1b}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{4.1c}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}. \tag{4.1d}$$

这里，E 为电场强度，D 为电通量密度，二者满足 D = εE，ε 为介电率；H 为磁场强度，B 为磁通密度，二者满足 B = μH，μ 为磁导率；ρ_v 为体电荷密度，J 为电流密度。E、D、B、H 已在第1-3节介绍，ρ_v 和 J 将在第4-2节讨论。麦克斯韦方程组适用于包括自由空间（真空）在内的各类介质。一般而言，上述量均可依赖空间位置与时间 t；为使记号简洁，除上下文需要外，不显式写出 E(x,y,z,t) 等函数的自变量。麦克斯韦在1873年出版的经典著作中系统阐述了这些方程，建立了统一的电磁理论。这些方程综合了库仑、高斯、安培、法拉第等人的实验成果，不仅描述电荷与电场、电流与磁场之间的联系，也描述电场与磁场之间的双向耦合。麦克斯韦方程组及相应的辅助关系共同构成电磁理论的基础。

► 在静态条件下，麦克斯韦方程组中出现的所有量都不是时间的函数（即 $\partial/\partial t = 0$）。当所有电荷永久固定在空间中时，就会发生这种情况。如果它们移动，它们会以稳定的速率移动，因此 ρ_v 和 J 在时间上恒定。◄

在这种情况下，方程中B和D的时间导数。 (4.1b) 和 (4.1d) 消失，麦克斯韦方程组简化为以下对。

静电学

$$\nabla \cdot \mathbf{D} = \rho_v, \tag{4.2a}$$

$$\nabla \times \mathbf{E} = 0. \tag{4.2b}$$

静磁学

$$\nabla \cdot \mathbf{B} = 0, \tag{4.3a}$$

$$\nabla \times \mathbf{H} = \mathbf{J}. \tag{4.3b}$$

麦克斯韦的四个方程分成两对非耦合方程，第一对仅涉及电场以及磁通 E 和 D，第二对仅涉及磁场以及磁通 H 和 B。

► 电场和磁场在静态条件下会解耦。◄

只要电荷和电流的空间分布随时间保持恒定，这使我们能够将电和磁作为两种不同且独立的现象来研究。我们将静态条件下的电磁现象的研究分别称为静电学和静磁学。静电学是本章的主题，我们在第五章中学习静磁学。通过研究静电和静磁现象获得的经验对于处理后续章节中涉及时变场、电荷密度和电流的更复杂的材料将证明是非常宝贵的。

我们研究静电不仅是作为研究时变领域的前奏，而且还因为它本身就是一个重要的领域。许多电子设备和系统都是基于静电原理。它们包括 X 射线机、示波器、喷墨静电打印机、液晶显示器、复印机、微机电开关和加速度计以及许多固态控制设备。静电原理还指导医疗诊断传感器的设计，例如记录心脏泵血模式的心电图和记录大脑活动的脑电图，以及众多工业应用的开发。

4-2 电荷和电流分布

在电磁学中，我们遇到各种形式的电荷分布。当运动时，这些电荷分布就构成了电流分布。电荷和电流可以分布在一定体积的空间、整个表面或沿着一条线。

4-2.1 Charge Densities

At the atomic scale, the charge distribution in a material is discrete, meaning that charge exists only where electrons and nuclei are and nowhere else. In electromagnetics, we usually are interested in studying phenomena at a much larger scale, typically three or more orders of magnitude greater than the spacing between adjacent atoms. At such a macroscopic scale, we can disregard the discontinuous nature of the charge distribution and treat the net charge contained in an elemental volume $\Delta\upsilon$ as if it were uniformly distributed within. Accordingly, we define the ***volume charge density*** ρ_v as

$$\rho_v = \lim_{\Delta\upsilon\to 0}\frac{\Delta q}{\Delta\upsilon} = \frac{dq}{d\upsilon} \qquad (\text{C/m}^3), \tag{4.4}$$

where Δq is the charge contained in $\Delta\upsilon$. In general, ρ_v depends on spatial location (x,y,z) and t; thus, $\rho_v = \rho_v(x,y,z,t)$. Physically, ρ_v represents the average charge per unit volume for a volume $\Delta\upsilon$ centered at (x,y,z) with $\Delta\upsilon$ being large enough to contain a large number of atoms, yet it is small enough to be regarded as a point at the macroscopic scale under consideration. The variation of ρ_v with spatial location is called its ***spatial distribution*** (or simply its ***distribution***). The total charge contained in volume υ is

$$Q = \int_\upsilon \rho_v\, d\upsilon \qquad (\text{C}). \tag{4.5}$$

In some cases, particularly when dealing with conductors, electric charge may be distributed across the surface of a material, where the quantity of interest is the ***surface charge density*** ρ_s, which is defined as

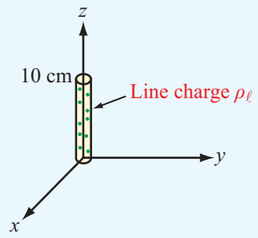
$$\rho_s = \lim_{\Delta s\to 0}\frac{\Delta q}{\Delta s} = \frac{dq}{ds} \qquad (\text{C/m}^2), \tag{4.6}$$

where Δq is the charge present across an elemental surface area Δs. Similarly, if the charge is, for all practical purposes, confined to a line, which need not be straight, we characterize its distribution in terms of the ***line charge density*** ρ_ℓ, defined as

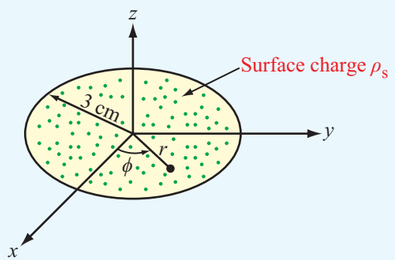
$$\rho_\ell = \lim_{\Delta l\to 0}\frac{\Delta q}{\Delta l} = \frac{dq}{dl} \qquad (\text{C/m}). \tag{4.7}$$

Example 4-1: Line Charge Distribution

Calculate the total charge Q contained in a cylindrical tube oriented along the z axis, as shown in **Fig. 4-1(a)**. The line charge density is $\rho_\ell = 2z$, where z is the distance in meters from the bottom end of the tube. The tube length is 10 cm.



(a) Line charge distribution



(b) Surface charge distribution

Figure 4-1 Charge distributions for Examples 4-1 and 4-2.

Solution: The total charge Q is

$$Q = \int_0^{0.1}\rho_\ell\, dz = \int_0^{0.1} 2z\, dz = z^2\Big|_0^{0.1} = 10^{-2}\ \text{C}.$$

Example 4-2: Surface Charge Distribution

The circular disk of electric charge shown in **Fig. 4-1(b)** is characterized by an azimuthally symmetric surface charge density that increases linearly with r from zero at the center to 6 C/m^2 at $r = 3$ cm. Find the total charge present on the disk surface.

Solution: Since ρ_s is symmetrical with respect to the azimuth angle ϕ, it depends only on r and is given by

$$\rho_s = \frac{6r}{3\times 10^{-2}} = 2\times 10^2 r \qquad (\text{C/m}^2),$$

where r is in meters. In polar coordinates, an elemental area is $ds = r\, dr\, d\phi$, and for the disk shown in **Fig. 4-1(b)**, the

4-2.1 电荷密度

在原子尺度上，材料中的电荷分布是离散的，这意味着电荷仅存在于电子和原子核所在的地方，而不存在于其他地方。在电磁学中，我们通常有兴趣研究更大尺度的现象，通常比相邻原子之间的间距大三个或更多数量级。在这样的宏观尺度上，我们可以忽略电荷分布的不连续性质，并将元素体积 $\Delta\upsilon$ 中包含的净电荷视为均匀分布。因此，我们将体积电荷密度 ρ_v 定义为

$$\rho_v = \lim_{\Delta\upsilon\to 0}\frac{\Delta q}{\Delta\upsilon} = \frac{dq}{d\upsilon} \qquad (\text{C/m}^3), \tag{4.4}$$

其中 Δq 是 $\Delta\upsilon$ 中包含的电荷。一般来说，ρ_v 取决于空间位置 (x, y, z) 和 t；因此，$\rho_v = \rho_v(x, y, z, t)$。物理上，$\rho_v$ 表示以 (x, y, z) 为中心的体积 $\Delta\upsilon$ 的每单位体积的平均电荷，其中 $\Delta\upsilon$ 大到足以包含大量原子，但又小到足以被视为所考虑的宏观尺度上的一个点。ρ_v随空间位置的变化称为其空间分布（或简称其分布）。体积 υ 中包含的总电荷为

$$Q = \int_\upsilon \rho_v\, d\upsilon \qquad (\text{C}). \tag{4.5}$$

在某些情况下，特别是在处理导体时，电荷可能分布在材料的表面，其中感兴趣的量是表面电荷密度 ρ_s，其定义为

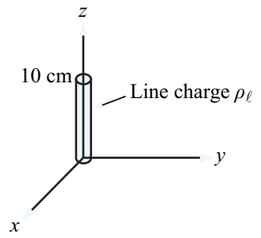
$$\rho_s = \lim_{\Delta s\to 0}\frac{\Delta q}{\Delta s} = \frac{dq}{ds} \qquad (\text{C/m}^2), \tag{4.6}$$

其中 Δq 是元素表面积 Δs 上存在的电荷。类似地，如果出于所有实际目的，电荷被限制在一条线上，而这条线不必是直的，我们就用线电荷密度 ρ_ℓ 来表征其分布，定义为

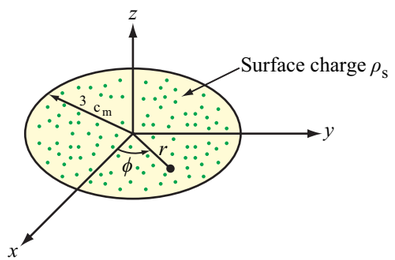
$$\rho_\ell = \lim_{\Delta l\to 0}\frac{\Delta q}{\Delta l} = \frac{dq}{dl} \qquad (\text{C/m}). \tag{4.7}$$

例 4-1：线路电荷分布

计算沿 z 轴定向的圆柱管中所含的总电荷 Q，如图 4-1（a）所示。线电荷密度为 $\rho_\ell = 2z$，其中 z 是距管底端的距离（以米为单位）。管子长度为 10 厘米。



(a) Line charge distribution



(b) Surface charge distribution

图 4-1 示例 4-1 和 4-2 的电荷分布。

解：总电荷Q为

$$Q = \int_0^{0.1}\rho_\ell\, dz = \int_0^{0.1} 2z\, dz = z^2\Big|_0^{0.1} = 10^{-2}\ \text{C}.$$

例 4-2：表面电荷分布

图 4-1(b) 所示的电荷圆盘的特征是方位对称的表面电荷密度，随着 r 从中心的 0 线性增加到 r = 3 cm 处的 6 C/m^2。找出磁盘表面上存在的总电荷。

解：由于 ρ_s 相对于方位角 ϕ 对称，因此它仅取决于 r，由下式给出

$$\rho_s = \frac{6r}{3\times 10^{-2}} = 2\times 10^2 r \qquad (\text{C/m}^2),$$

其中 r 的单位是米。在极坐标中，一个基本区域为 ds = r dr dϕ，对于图 4-1(b)所示的圆盘，

limits of integration are from 0 to 2π (rad) for ϕ and from 0 to 3×10^{-2} m for r. Hence,

$$Q = \int_S \rho_s \, ds = \int_{\phi=0}^{2\pi} \int_{r=0}^{3\times10^{-2}} (2 \times 10^2 r) r \, dr \, d\phi$$

$$= 2\pi \times 2 \times 10^2 \, \frac{r^3}{3}\Big|_0^{3\times10^{-2}} = 11.31 \quad \text{(mC)}.$$

Exercise 4-1: A square plate residing in the x–y plane is situated in the space defined by -3 m $\le x \le 3$ m and -3 m $\le y \le 3$ m. Find the total charge on the plate if the surface charge density is $\rho_s = 4y^2$ (μC/m^2).

Answer: $Q = 0.432$ (mC). (See EM.)

Exercise 4-2: A thick spherical shell centered at the origin extends between $R = 2$ cm and $R = 3$ cm. If the volume charge density is $\rho_v = 3R \times 10^{-4}$ (C/m^3), find the total charge contained in the shell.

Answer: $Q = 0.61$ (nC). (See EM.)

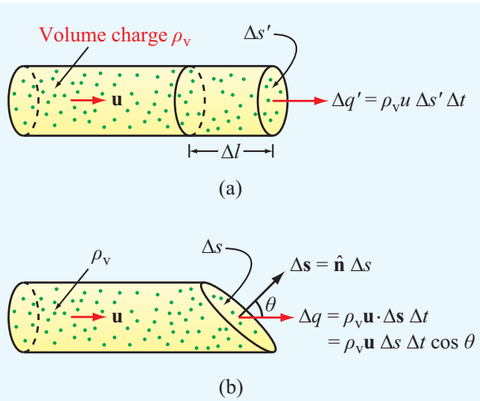


Figure 4-2 Charges with velocity **u** moving through a cross section $\Delta s'$ in (a) and Δs in (b).

4-2.2 Current Density

Consider a tube with volume charge density ρ_v (Fig. 4-2(a)). The charges in the tube move with velocity **u** along the tube axis. Over a period Δt, the charges move a distance $\Delta l = u \, \Delta t$. The amount of charge that crosses the tube's cross-sectional surface $\Delta s'$ in time Δt is therefore

$$\Delta q' = \rho_v \, \Delta \mathcal{V} = \rho_v \, \Delta l \, \Delta s' = \rho_v u \, \Delta s' \, \Delta t. \tag{4.8}$$

Now consider the more general case where the charges are flowing through a surface Δs with normal $\hat{\mathbf{n}}$ not necessarily parallel to **u** (Fig. 4-2(b)). In this case, the amount of charge Δq flowing through Δs is

$$\Delta q = \rho_v \mathbf{u} \cdot \Delta \mathbf{s} \, \Delta t, \tag{4.9}$$

where $\Delta \mathbf{s} = \hat{\mathbf{n}} \, \Delta s$ and the corresponding total current flowing in the tube is

$$\Delta I = \frac{\Delta q}{\Delta t} = \rho_v \mathbf{u} \cdot \Delta \mathbf{s} = \mathbf{J} \cdot \Delta \mathbf{s}. \tag{4.10}$$

Here

$$\mathbf{J} = \rho_v \mathbf{u} \qquad (\text{A/m}^2) \tag{4.11}$$

is defined as the ***current density*** in ampere per square meter. Generalizing to an arbitrary surface S, the total current flowing through it is

$$I = \int_S \mathbf{J} \cdot d\mathbf{s} \qquad \text{(A)}. \tag{4.12}$$

► When a current is due to the actual movement of electrically charged matter, it is called a ***convection current***, and **J** is called a ***convection current density***. ◄

A wind-driven charged cloud, for example, gives rise to a convection current. In some cases, the charged matter constituting the convection current consists solely of charged particles, such as the electron beam of a scanning electron microscope or the ion beam of a plasma propulsion system.

When a current is due to the movement of charged particles relative to their host material, **J** is called a ***conduction current density***. In a metal wire, for example, there are equal amounts of positive charges (in atomic nuclei) and negative charges (in the electron shells of the atoms). None of the positive charges and few of the negative charges can move; only those electrons in the outermost electron shells of the atoms can be pushed from one atom to the next if a voltage is applied across the ends of the wire.

► This movement of electrons from atom to atom constitutes a ***conduction current***. The electrons that emerge from the wire are not necessarily the same electrons that entered the wire at the other end. ◄

ϕ 的积分限制为 0 到 2π (rad)，r 的积分限制为 0 到 3×10^{-2} m。因此，

$$Q = \int_S \rho_s \, ds = \int_{\phi=0}^{2\pi} \int_{r=0}^{3\times10^{-2}} (2 \times 10^2 r) r \, dr \, d\phi$$

$$= 2\pi \times 2 \times 10^2 \, \frac{r^3}{3}\Big|_0^{3\times10^{-2}} = 11.31 \quad \text{(mC)}.$$

练习 4-1：位于 x-y 平面上的方形板位于由 -3 m $\le$ x ≤ 3 m 和 -3 m $\le$ y ≤ 3 m 定义的空间中。如果表面电荷密度为 $\rho_s = 4y^2$ (μ C/m^2)，求极板上的总电荷。

答案：Q = 0.432 (mC)。 （参见EM。）

练习 4-2：以原点为中心的厚球壳在 R = 2 cm 和 R = 3 cm 之间延伸。如果体积电荷密度为 $\rho_v = 3R \times 10^{-4}$ (C/m^3)，求壳层中包含的总电荷。

答案：Q = 0.61 (nC)。 （参见EM。）

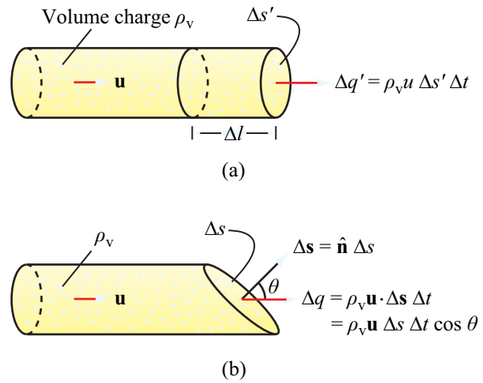


图 4-2 电荷以速度 u 穿过 (a) 中的横截面 $\Delta s'$ 和 (b) 中的 Δs 横截面。

4-2.2 电流密度

考虑具有体积电荷密度 ρ_v (图 1 的管子。 4-2(a))。管内的电荷沿管轴以速度 u 移动。在 Δt 期间，电荷移动距离 $\Delta l = u \, \Delta t$。因此，在时间 Δt 内穿过管横截面 $\Delta s'$ 的电荷量为

$$\Delta q' = \rho_v \, \Delta \mathcal{V} = \rho_v \, \Delta l \, \Delta s' = \rho_v u \, \Delta s' \, \Delta t. \tag{4.8}$$

现在考虑更一般的情况，其中电荷流过表面 Δs，其法线 $\hat{}$ n 不一定平行于 u（图 4-2(b)）。在这种情况下，流过 Δs 的电荷量 Δq 为

$$\Delta q = \rho_v \mathbf{u} \cdot \Delta \mathbf{s} \, \Delta t, \tag{4.9}$$

其中 $\Delta s = \hat{n} \, \Delta s$ 和相应的管中流动的总电流为

$$\Delta I = \frac{\Delta q}{\Delta t} = \rho_v \mathbf{u} \cdot \Delta \mathbf{s} = \mathbf{J} \cdot \Delta \mathbf{s}. \tag{4.10}$$

这里

$$\mathbf{J} = \rho_v \mathbf{u} \qquad (\text{A/m}^2) \tag{4.11}$$

定义为电流密度，单位为安培每平方米。推广到任意表面 S，流过的总电流

$$I = \int_S \mathbf{J} \cdot d\mathbf{s} \qquad \text{(A)}. \tag{4.12}$$

► 当电流是由于带电物质的实际运动而产生时，称为对流电流，J 称为对流电流密度。 ◄

例如，风驱动的带电云会产生对流。在一些情况下，构成对流的带电物质仅由带电粒子组成，例如扫描电子显微镜的电子束或等离子体推进系统的离子束。

当电流是由于带电粒子相对于其主体材料的运动而产生时，J 称为传导电流密度。例如，在金属线中，有等量的正电荷（在原子核中）和负电荷（在原子的电子壳层中）。正电荷和负电荷都不能移动；如果在导线两端施加电压，则只有原子最外层电子壳层中的电子才能从一个原子推到下一个原子。

► 电子从一个原子到另一个原子的运动构成了传导电流。从导线中出现的电子不一定与在另一端进入导线的电子相同。 ◄